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2

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Question 1.

What model of teaching will this course follow?

Peerless model

Flipped model throughout

Regular lectures throughout

Part flipped, part regular

CHAT



1

2

3

4

Question 2.

What is the attendance policy in CS348 and CS378 ?

Minimum 75% needed, else DX

100% needed, else DX

50% needed, else one grade down

All classes are optional as videos and ChatGPT are available

CHAT

Screenshot saved



1

2

3

4

Question 3.

When will there be a SAFE quiz in the CS348 course?

A randomly chosen 5 min during each class

First 5 min of each class

Last 5 min of each class

Only 4 times in the semester

CHAT



1

2

3

4

Question 4.

What does brain drain refer to in the context of the information age?

Indian graduates seeking job abroad

Cognitive ability being affected by presence of your network-connected smart-phone nearby

Electro-magnetic waves affecting your concentration

Wastage of time studying a computer networks course

CHAT



1

2

3

4

Question 1.

Which of the following are components of delay in a network? Select ALL that apply.

Queueing

Propagation

Transmission

Processing

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1

2

3

4

Question 2.

What is the role of the physical layer in a computer network?

To manufacture cables, links, wires

To transform bits to energy and vice versa

To transform alphabets to bits and vice versa

To provide reliability in transmission

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1

2

3

4

Question 3.

Which ONE of the following is the layer which is responsible for reaching packets from the source host to the destination host?

Medium Access Control

Transport

Presentation

Network

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1

2

3

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Question 4.

What does encapsulation refer to in multi-layer communication?

The addition of a header as the packet moves down the protocol stack

The addition of a header as the packet moves up the protocol stack

The conversion of alphabets to bits

The conversion of bits to alphabets

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1

2

3

4

Question 1.

Which among the following links gives the highest data rate?

Twisted Pair

Coaxial Cable

Fiber Optics

Wireless

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1

2

3

4

Question 2.

Which of the following is true of signals and bandwidth? Select all that apply.

Aperiodic signals have discrete frequencies in the freq domain

Nyquist Rate considers the effect of noise

Shannon's theorem gives the lower bound of the capacity of a link

In Shannon's theorem, if $R > C$, the probability of error increases without bound

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1

2

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Question 3.

Which of the following is true of Manchester encoding? Select all that apply.

Suffers from baseline wander

Has a significant DC component

Self-clocking

More efficient than NRZI in terms of data rate

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1

2

3

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Question 4.

What is the efficiency of 4B/5B in percentage?
(round off to integer)

ANSWER:

80.0 - 80.0

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1

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3

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Question 1.

In the 4B/5B encoding scheme, what is the maximum number of consecutive zeroes possible in the encoded bit pattern?

Answer -1 if it is not possible to determine.

Answer 9999 if the answer is infinity.

ANSWER:

3 - 3

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1

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Question 2.

Which of the following are types of modulation schemes? Select all that apply.

Line encoding

Baseband modulation

Shannon scheme

Fourier scheme

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1

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Question 3.

What is the fundamental frequency of a square wave of period 1 micro-second? Give your answer in MHz. Answer -1 if it is not possible to determine. Answer 9999 if the answer is infinity.

ANSWER:

1 - 1

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Question 4.

Nyquist criterion relates which of the following two quantities?

The achievable bit rate and the link bandwidth

The link bandwidth and the bit error rate

The bit error rate and the packet error rate

The packet error rate and the maximum link length

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1

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Question 1.

Which of the following are advantages of 4B/5B encoding compared to Manchester encoding? Select all that apply.

4B/5B can be used for wireless links,
Manchester cannot be

**4B/5B is more bandwidth efficient
compared to Manchester encoding**

4B/5B handles baseline wander,
Manchester does not

4B/5B handles sync loss (clock drift),
Manchester does not

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1

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Question 2.

A random data signal with a 5Kbps bit rate, encoded using Manchester encoding has a first null bandwidth of ____ KHz. Give your answer rounded to the nearest integer.

ANSWER:

10 - 10

CHAT



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Question 3.

The link has a bandwidth of 5MHz and a signal-to-noise ratio (SNR) of 1000. What is its Shannon capacity limit? Answer in Mbps, rounded to the nearest integer.

ANSWER:

50 - 50

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Question 4.

Which ONE of the following encoding/modulation is UNSUITABLE for wireless links?

Amplitude modulation

NRZ (Non-Return to Zero)

Encoding based on phase of wave

Frequency modulation

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1

2

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Question 1.

The unit of information carried at the link layer is ____.

Bit

Byte

Frame

Packet

Hop

CHAT



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Question 2.

Which of the following are ways of recovering from bit errors over a link?
Select all that apply.

Medium Access Control protocol

Redundant bits sent along with original transmission

Retransmission of frames in error

Use a special byte such as DLE (Data Link Escape)

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1

2

3

4

Question 3.

What is the Hamming distance of a scheme which sends every bit in duplicate?

ANSWER:

2 - 2

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1

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3

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Question 4.

Ethernet's CRC (Cyclic Redundancy Check) is used for:

Error detection

Error correction

Retransmission

Frame separation

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Question 1.

Which ONE of the following functionalities is OPTIONAL at the link layer?

Framing

Modulation

Error recovery

Multiplexing

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Question 2.

In which of the following conditions is retransmission of errored frames better than using redundant bits in every frame? Select all that apply.

Low bit error rate

Low propagation delay

Satellite links

None of the other options

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1

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Question 3.

What is the Hamming distance of a scheme which sends the complement of the data bit as a redundant information along with the data bit?

ANSWER:

2 - 2

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Question 4.

Which ONE of the following is an advantage of CRC over 2-D parity?

It can correct single-bit errors

It can correct 2-bit errors

It has much lower bit overhead

It can work over wired links as well as wireless links

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Question 1.

Which of the following are sub-layers of the link layer?

LLC and MAC

FEC and ARQ

Ethernet and WiFi

PPP and Framing

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1

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Question 2.

In which of the following conditions is transmission of redundant bits in every frame better than retransmission of errored frames? Select all that apply.

High propagation delay

High noise scenario

Fiber optic cables

None of the other options

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Question 3.

An error correction scheme sends four copies of each data bit. How many bit errors can this scheme correct?

ANSWER:

1 - 1

CHAT



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Question 4.

Which ONE of the following is used in Internet checksum algorithm?

8-bit 2's complement addition

16-bit 2's complement addition

16-bit 1's complement addition

8-bit 1's complement addition

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Question 1.

A link layer protocol is designed to be reliable, but using only NACKs. Choose the correct option below about this protocol:

it does not need data sequence numbers

it is correct in all possible packet error scenarios but not always efficient

its correctness cannot be guaranteed in all possible packet error scenarios

it does not need NACK sequence numbers

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Question 2.

How MANY different sequence numbers are needed (i.e. the size of the sequence number space) for correct operation of the sliding window protocol, for an Earth-to-satellite link with propagation delay of 1sec and transmission delay of 1ms? Answer -1 if it is not possible to determine based on the given information.

ANSWER:

2 - 2

The answer -1 is also being considered as the question mentions that it is a sliding window protocol and we don't know the SWS and RWS values for the protocol. If we consider it to be a stop and wait protocol i.e. $SWS = RWS = 1$, we get 2 as the answer.

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Question 3.

A link layer reliability protocol has $RWS=1$ and $SWS=6$. Which of the following is/are true about the protocol? Select ALL that apply.

It is called go-back-N

It is equivalent to a protocol with $SWS=RWS=1$

It is equivalent to a protocol with $SWS=RWS=6$

None of the other options

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Question 4.

The Sorcerer's Apprentice bug in the stop-and-wait protocol is caused by:

loss of 2 consecutive ACKs

loss of 2 consecutive NACKs

premature timeout

loss of 2 consecutive data packets

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Question 1.

What is another name for the stop-and-wait protocol?

Alternating bit protocol

Step-and-move protocol

Go-back-N protocol

WiFi link protocol

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Question 2.

What is the utilization (in percentage) of stop and wait over a link with 4 Mbps speed and 2 ms one-way propagation delay with a packet size of 2000 Bytes?

ANSWER:

50.0 - 50.0

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answer explanation can be mentioned here



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Question 3.

Which among the following is/are required functionality (not optional) in link layer RDT protocols? Select all the apply.

Forward Error Correction

Error Detection

Sequence Number

ACK

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Question 4.

Which ONE of the following is always true about the go-back-N protocol?

RWS=1

SWS=1

SWS=RWS

None of the other options

CHAT



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Question 1.

Which of the following is/are true about NACK-only reliability protocols at the link layer? Select all that apply.

They cannot always guarantee correct operation

A long silence from the sender could mean a long delay in packet loss recovery

Loss of more than 1 consecutive packet implies incorrect operation

Loss of NACK packets implies incorrect operation

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Question 2.

Which of the following are possible SWS values for the stop-and-wait protocol?
Select all that apply.

0

1

2

10

CHAT



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Question 3.

How many bits are **required** to represent the sequence number field in the stop-and-wait protocol? Answer -1 if you cannot answer based on the given information.

ANSWER:

1 - 1

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Question 4.

Which ONE of the following is an advantage of the go-back-N protocol?

It does not require a sequence number field in the data packets

It does not require a sequence number field in the ACK packets

The receiver implementation is very simple

The sender implementation is very simple

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Question 1.

Which ONE of the following MAC schemes is best at low load?

Time division multiple access

Frequency division multiple access

Carrier sense multiple access

Code division multiple access

CHAT



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Question 2.

Which of the following is/are true about collision detection at the MAC protocol layer?

It is possible in Ethernet and is used in a hub-based star topology

It is possible in Ethernet and it used to detect collisions across multiple ports of a switch

It is possible in WiFi in Infrastructure BSS mode

It is not possible in WiFi

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Question 3.

Which ONE of the following is true about WiFi's MAC protocol?

It uses CSMA/CA only in ad-hoc mode but not in Infrastructure BSS mode

It uses CSMA/CA in both ad-hoc and Infrastructure BSS modes

It uses CSMA/CA only in the client-to-AP direction

It uses CSMA/CA in only the AP-to-client direction

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Question 4.

Which ONE of the following problems is faced in WiFi's MAC protocol operation?

Hidden-terminal problem

Spurious timeout problem

Long-frame problem

Duplicate-frame problem

CHAT



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Question 1.

Which ONE of the following is NOT a class of MAC protocols?

Time division multiple access

Code division multiple access

Polling

Staffing

CHAT



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Question 2.

Which of the following is/are true about WiFi's MAC protocol?

It includes a random backoff prior to sending a frame

It optionally includes the RTS/CTS mechanism to address the hidden terminal problem

It lowers the backoff after a collision to improve the probability of successful transmission

It employs a TDMA scheme controlled by the Access Point

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Question 3.

Which ONE of the following is true about the SLOT parameter in WiFi's MAC protocol?

It needs to be at least as large as the maximum expected RTT

It needs to be at least as large as the transmission time of the largest packet

It needs to be at least as large as the transmission time of the smallest packet

It needs to be at most as large as the transmission time of the smallest packet

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Question 4.

Which ONE of the following is true about Ethernet hubs?

They allow the range of the Ethernet to be extended indefinitely

They can only connect Ethernet wires of the same speed

They help avoid collisions between multiple Ethernet hosts

They help improve reliability by sending immediate ACKs

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Question 1.

Which ONE of the following is an ADVANTAGE of circuit switching compared to packet switching?

Better fault tolerance

Can support more users for the same available capacity

Can reserve resources for a flow of packets

Lower flow setup delay

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Question 2.

Which of the following are types of switching? Select all that apply.

Virtual circuit switching

Backpressure switching

Datagram switching

Envelope switching

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Question 3.

What does an Ethernet switched network form, to enable efficient switching?

Minimum spanning tree

Two-connected graph

Tree rooted at min-ID bridge

Hamilton cycle

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Screenshot saved



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Question 4.

Ethernet bridges learn the path to a host from _____.

source address of a frame

destination address of a frame

master broadcast

manufacturer's NIC entry

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